

User Guide

Beyond Mean-Variance Portfolio Optimiser

With Derivatives & Structured Products — A Mental Accounts Framework

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sami-jeddou-behavioral-portfolio-optimizer.streamlit.app

Home

The landing screen — your starting point: the six tools, with the reference shelf (About · Glossary · User Guide) below.

Portfolio Optimisation *with Derivatives & Structured Products*

Beyond Mean-Variance

Mental Accounting Framework

TOOLS



Risk Profile

13-question Grabie-Lyttan scale that sets your simulation's risk parameters (μ , σ , λ).

[find your risk level](#)



Ticker Analytics

Enter a ticker for its price, return, risk level and key ratios, each with a plain-language explanation.

[stocks · ETFs · indices](#)



Grid Portfolio Optimiser

Exact grid engine on the Des-Solman states — VaR, thesis-fairful ES and rigorous-ES, with derivatives.

[best for small portfolio](#)



Scalable Portfolio Optimiser

Monte-Carlo scenarios + α -CvAR linear program — scales to large, multi-derivative portfolios.

[best for many assets · beta](#)



Backtest

Out-of-sample walk-forward of the Optimiser's portfolios, derivative marked to market.

[% realised alpha & beta vs a benchmark](#)

[performance check](#)



Live Portfolio

Build a portfolio, save it, and track its return, risk level, alpha and beta over time.

[% stress-test against historical crisis](#)

[build · save · track](#)

REFERENCE



About

Methods, framework and research.



Glossary AI-POWERED

VaR, ES, α -CvAR, copulas — plus natural-language Q&A.



User Guide

Step-by-step tour of the app (PDF download).

Built by Sami Jeddou · [About & contact](#) · [Connect on LinkedIn](#)

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Home — choose a tool



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— VaR, thesis-ES and rigorous-ES, with
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Scalable Portfolio Optimiser

Monte-Carlo scenarios + α -CVaR linear
program — scales to large, multi-derivative
portfolios.



Backtest

Out-of-sample walk-forward — expected vs
realised, plus realised alpha & beta.

Reference & navigation



About:

Methods, the mental-accounting framework and the research behind the app.



Glossary:

Plain-language, AI-powered definitions — VaR, ES, α -CVaR, copulas, alpha & beta.



Tip:

Each tile opens a tool; the Grid Portfolio Optimiser is the place to start.

Grid Portfolio Optimiser — overview



Interactive Frontiers

MV vs Behavioural vs Derivative frontier — visualised side by side on a single chart



Three Portfolios

No derivative / With derivative / Same risk — three perspectives for each optimisation run



AI-Powered

On-demand explanations for every parameter, constraint, derivative type, and result

Portfolio perspectives (0-3)

0

Portfolio (0) — Markowitz MV optimum:

Classical mean-variance solution; Portfolio (1) coincides with it.

1

Portfolio (1) — No derivative:

Equivalent to MV optimum — confirms MVT/MAT equivalence. Most classical optimisers stop here.

2

Portfolio (2) — With derivative:

Same implied λ and downside constraint (H, α or H, α, L) — shows return uplift from derivatives.

3

Portfolio (3) — Same risk as (1):

Interpolated at $P(1)$ std dev — direct comparison of return gain at equivalent risk (indicative).

Step 1 — Portfolio Data

1

PORTFOLIO DATA

🔮 AI-powered: How these inputs are built

☒ Default (3-asset sample case)

☐ Live market data (Yahoo Finance)

☐ Enter manually

☐ Upload CSV

✓ Default base case loaded — Means: 5%, 10%, 25% | Std devs: 5%, 20%, 50%

Default

Das & Statman (2009) base case — 3 securities, pre-calibrated. Reproduces thesis results.

Live market data

Any global ticker from Yahoo Finance — equities, ETFs, indices, crypto (BTC, ETH...)

Enter manually

Enter means, std devs, and correlation matrix directly. Supports 2–10 securities.

Upload CSV

First col = dates, remaining cols = asset prices. Returns computed automatically.

Important: with Live market data or CSV, click 'Fetch data' to load it before running the optimiser — it will not run on un-fetched tickers.

Step 2 — Derivative / Structured Product

2 DERIVATIVE / STRUCTURED PRODUCT

None — primary securities only

None — primary securities only

Put option

Call option

Safety collar (long put + short call)

Aggressive collar (long call + short ...)

Straddle (long call + long put)

Strangle (long call + long put, diff s...)

2 DERIVATIVE / STRUCTURED PRODUCT

Straddle (long call + long put)

Payoff diagram

AI-powered: What is this instrument?

Underlying security
Sec 3 — High risk

Volatility (annualised %)
50.0

Default: 50.0% (std dev of Sec 3 — High risk).
Override for implied vol.

Risk-free rate (%)
3.0

Maturity (years)
1.00

Strike (fraction of spot)
0.78

Available instruments

- ▶ Vanilla options — put, call
- ▶ Collars — safety, aggressive
- ▶ Volatility — straddle, strangle
- ▶ Spreads — bull call, bear put
- ▶ Butterfly & condor — long butterfly, call condor
- ▶ Capital-guaranteed note — capped or uncapped
- ▶ Barrier-M note & reverse convertible
- ▶ Discount & outperformance certificates

? Select 'None' for no derivatives, or 'Custom' to build your own (calls, puts, digitals, ZCBs).

Step 3 — Mental-Account Constraint

3

MENTAL-ACCOUNT
CONSTRAINT

AI-powered: What is the VaR constraint?

Constraint type

☒ VaR — Value at Risk
 ☐ ES — Expected Shortfall
 ☐ ES — Rigorous (beyond thesis)

Threshold H (%)

-40

-10

-1

Range extended to -40% to accommodate highly volatile assets (e.g. cryptocurrencies, emerging market equities).

Max shortfall probability α (%)

1

5

15

VaR constraint: $P(\text{return} < H) \leq \alpha$

Implied risk-aversion $\lambda = 3.6958$

MV optimal at $\lambda=3.70 \equiv$ behavioural optimal at

H=-10%, $\alpha=5\%$

VaR — Value at Risk

$$P(\text{return} < H) \leq \alpha$$

Caps the probability of finishing below H at α .

ES — Expected Shortfall

$$E[\text{return} \mid \text{return} < H] \geq L$$

Floors the average tail loss below H at L (thesis).

ES — Rigorous (beyond thesis)

$$E[r \mid r < H] \geq L \text{ (enforced in the solve)}$$

Enforces the ES floor in the optimisation itself.

Implied λ is computed dynamically — adjust H and α to see the equivalent risk-aversion coefficient update in real time, confirming MVT/MAT equivalence.

Step 4 — Grid Resolution



Also available

▶▶ **Turbo — 4th resolution:** High-precision VaR accuracy in ~seconds (VaR-only, ≤ 4 assets, no derivative).

Rigorous ES: a fixed high-precision grid ($m=51$); chosen in the Risk-measure step.

⚡ Fast ($m=21$, $m'=15$)

🕒 ~1-2 min (or more depending on the portfolio size)

Quick preview and parameter exploration

⚖️ Standard ($m=35$, $m'=50$)

🕒 ~5-15 min (or more depending on the portfolio size)

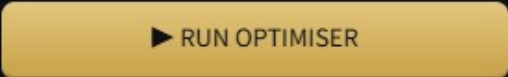
Good balance — close to thesis results

🎯 High precision ($m=51$, $m'=99$)

🕒 1-2 hrs (or more depending on the portfolio size)

Publication-quality replication of thesis results

Step 5 — Run the Optimiser



▶ RUN OPTIMISER

Computation steps

1. Mean-variance frontier — Markowitz sweep (instant)
2. Behavioural frontier (no derivative) — State space, Gaussian copula, grid search, COBYLA
3. Derivative frontier — Same steps extended with derivative payoff (only if derivative selected)
4. Chart rendering + Results computation

Reading the Chart

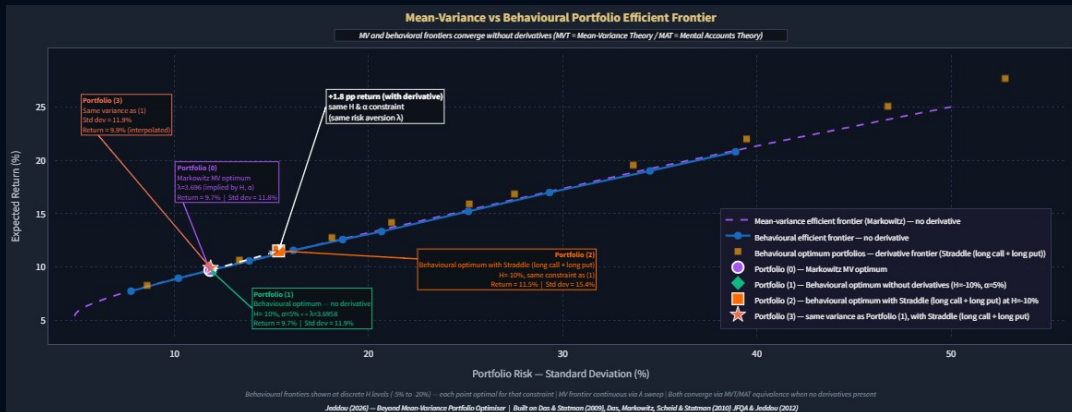


Chart legend

- Purple dashed**
Mean-variance efficient frontier (Markowitz)
- Blue dots**
Behavioural efficient frontier — no derivative
- Gold squares**
Behavioural optimum — derivative frontier
- Green diamond**
Portfolio (1) — no derivative optimum
- Orange square**
Portfolio (2) — with derivative, same λ
- Coral star**
Portfolio (3) — same variance as P(1)
- Magenta circle**
Portfolio (0) — Markowitz MV optimum

X-axis: Portfolio Risk (Standard Deviation %) | Y-axis: Expected Return (%)

Portfolio Results

Optimal portfolios with $H=-10\%$, $\alpha=5\%$ — implied $\lambda = 3.6958$

Summary — resulting portfolios

Portfolio	Expected return	Std deviation	Skewness	Δ vs (1)
• Portfolio (0) — Markowitz MV optimum (no derivative)	9.69%	11.83%	0.000	-0.00 pp
• Portfolio (1) — Behavioural optimum, no derivative	9.69%	11.87%	0.000	—
• Portfolio (2) — Behavioural optimum, with Straddle (long call + long put)	11.50%	15.37%	0.489	+1.81 pp
• Portfolio (3) — Same variance as Portfolio (1), with Straddle (long call + long put)	9.90%	11.87%	—	+0.21 pp

Δ vs (1) = expected-return gap relative to Portfolio (1). Portfolio (0) is a Gaussian mean-variance construct (skewness 0). Portfolio (3) is interpolated from the derivative frontier (indicative only).

Key metrics

Expected return: Annualised mean return of the optimal portfolio

Shortfall / ES: Expected loss in the tail (ES constraint value)

Std deviation: Annualised portfolio volatility (risk measure)

Portfolio weights: Allocation to each security and derivative

Skewness: Asymmetry of return distribution — positive = upside skew

PDF Export

↓ EXPORT & DOWNLOAD PDF REPORT

PDF report includes

- ✓ Cover page with run parameters
- ✓ Full efficient frontier chart
- ✓ Portfolio (1), (2), (3) metrics
- ✓ Weight allocation bar charts
- ✓ Legal disclaimer

📄 The PDF is generated in-browser — no data is sent externally. Click 'Download PDF report' after the simulation completes.

Scalable Portfolio Optimiser — Monte-Carlo + CVaR



Scenario engine

Samples thousands of joint return scenarios through a Gaussian or Student-t copula.



α -CVaR linear program

Maximises return subject to a CVaR (Expected-Shortfall) floor — Rockafellar–Uryasev.



Institutional scale

Cost grows linearly in the number of assets — dozens of holdings, several derivatives at once.

How to use it

1

Pick assets & scenarios:

Enter tickers and derivatives, choose the copula and the number of Monte-Carlo scenarios.

2

Set the ES floor:

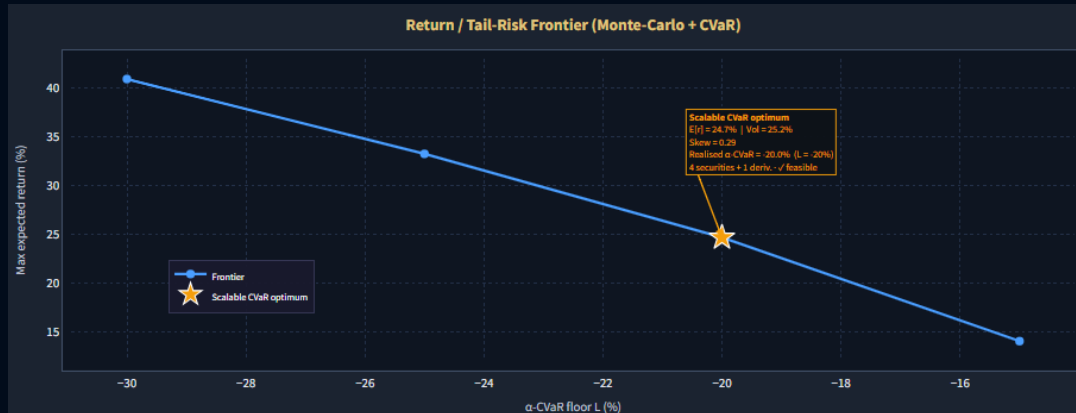
Choose the loss level L ; the engine sweeps the return / tail-risk frontier.

3

Read the optimum:

★ marks the chosen portfolio; weights, $E[r]$, realised ES, vol and skew are reported.

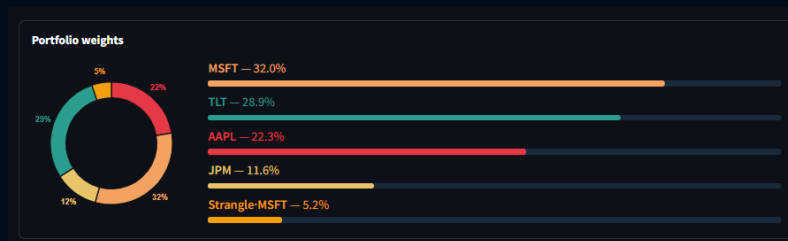
Reading the Scalable output



What you're seeing

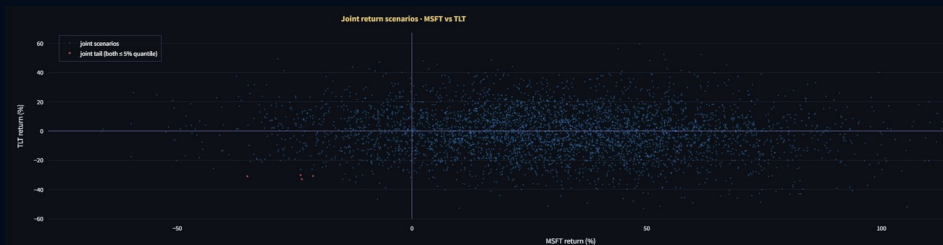
- Curve = the return / tail-risk frontier: the most expected return reachable for each Expected-Shortfall floor L.
- A tighter floor (toward -10%) allows less tail loss and lowers return; a looser one (toward -30%) raises it.
- The ★ is the optimum at your chosen floor (here L = -20%, E[r] = 24.6%).
- Built from Monte-Carlo scenarios solved by a CVaR linear program, so it scales to many assets.

Reading the Scalable output — detail



Portfolio return distribution

Simulated return of the chosen portfolio across every scenario. The shaded tail is the worst α -Choi, its average is the realised α -Choi, held at or above the floor L — so the shaded region is exactly what the Choi constraint controls.



Out-of-sample back-test



Two windows

Build weights on a construction window, then buy-and-hold them through a later evaluation window.



Marked to market

The derivative is repriced each step with Black-Scholes, giving a full realised return path.



Alpha & beta

Realised alpha, beta and R^2 of each holding and of the portfolio vs a chosen benchmark.

What you get

1

Expected vs realised:

Return, volatility and the loss-threshold outcome, for P1 (no derivative) and P2 (with).

2

Alpha / beta table:

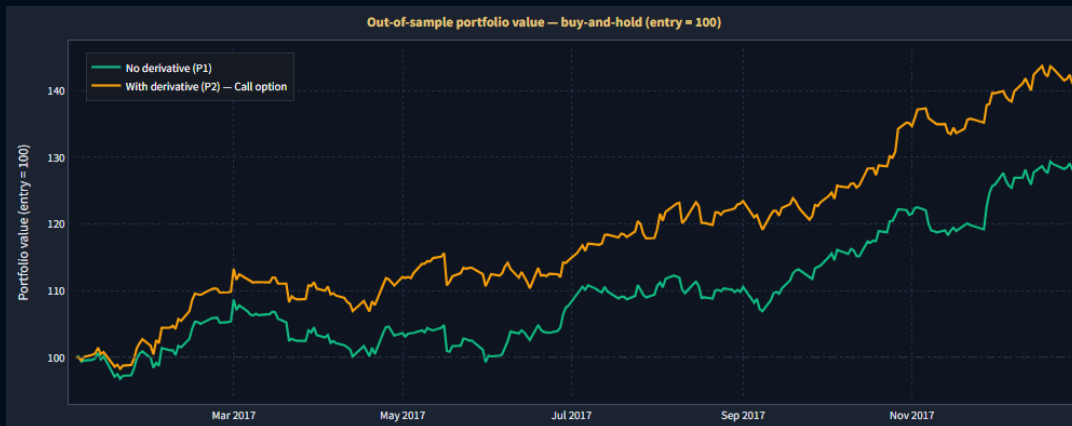
Per-security and portfolio beta (the regression slope) and annualised Jensen alpha, with R^2 .

3

Optional CAPM:

Enter an expected market return $E[R_m]$ to add a CAPM required return and an ex-ante alpha.

Reading the Backtest output



What you're seeing

- Out-of-sample value of two optimiser portfolios, rebased to 100 at entry and marked to market.
- Green = Portfolio (1), no derivative; orange = Portfolio (2), with the chosen derivative (a put).
- The derivative line cushions drawdowns and ends higher — the realised hedging benefit on unseen data.
- Realised alpha & beta vs a benchmark are computed from these paths (table beneath the chart in the app).

More tools — Profile, Ticker & Live Portfolio

Risk Profile

13-question Grable-Lytton scale → your tolerance band and the parameters H , α , L , with a dated history.

Ticker Analytics

Key figures and CFA-style ratios for any stock, ETF or index, each with a plain-language explanation.

Live Portfolio

Build and track a real book of securities and derivatives over time, marked to market.

How they fit together



Profile first:

size your risk before optimising.



Then build:

hold and monitor a real portfolio.



Tip:

the Live Portfolio is detailed next.

Live Portfolio — build & track

Holdings + derivatives

Add securities and option / structured-product positions (puts, calls, collars, spreads, CGN); each row is held from its entry date.

Marked to market

Derivatives are priced with real expiry on their underlying — the same engine as the Backtest.

Realised analytics

Return, volatility, Sharpe, drawdown, daily VaR/CVaR, and CAPM alpha, beta and R^2 vs your benchmark.

Setting it up



Total 100%:

securities + derivatives must sum to 100%.



Inception:

earliest entry date; positions accumulate.



View range:

narrow the analytics to any date window.

Live Portfolio — monitor & stress

Risk vs tolerance

Monitor against your limit with VaR (breaches $\leq \alpha$) or ES (tail $\geq L$); a time-varying mode uses a dated tolerance timeline.

Stress testing

Historical crises (2008–2022) projected forward, custom β -shock paths, and a parametric (vol $\times$, corr $\rightarrow$ 1) stress — with charts.

Save to your Drive

Sign in with Google to store portfolios in your own Drive (drive.file), or export a portable JSON file.

Saving & privacy



Bundled:

holdings, benchmark, risk settings, profile & timeline.



Privacy:

the app only touches files it created.

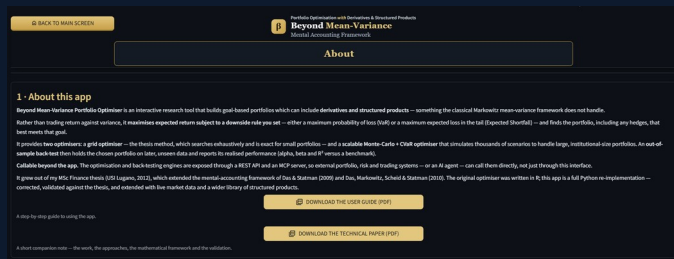


Tip:

sign in first, then load your saved portfolio.

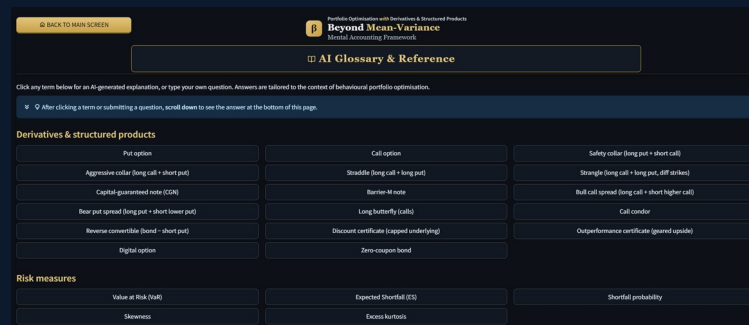
About & Glossary Tabs

About Tab



- ▶ Author background & contact
- ▶ App description & algorithm
- ▶ Academic references
- ▶ Legal disclaimer

Glossary Tab



- ▶ Click any term for AI explanation
- ▶ Covers derivatives, risk, portfolio theory
- ▶ Ask your own questions
- ▶ Tailored to behavioural portfolio context

Disclaimer & Contact

⚠ Legal Disclaimer

This application is based on the mental accounts portfolio optimisation framework of Das & Statman (2009) and Das, Markowitz, Scheid & Statman (2010), as extended in Jeddou (2012). It is provided for educational and research purposes only and does not constitute financial advice, investment recommendations, or a solicitation to buy or sell any financial instrument. Results are purely illustrative and should not be used as the basis for any investment decision.

Academic References

- ▶ Das, S. & Statman, M. (2009) — Beyond Mean-Variance: Portfolios with Derivatives and Non-Normal Returns in Mental Accounts
- ▶ Das, S., Markowitz, H., Scheid, J. & Statman, M. (2010) — Portfolio Optimization with Mental Accounts, JFQA Vol.45 No.2
- ▶ Jeddou, S. (2012) — Beyond Mean-Variance: Options and Structured Products in Behavioral Portfolios, MSc Finance, USI Lugano